Quadratic and cubic expressions

Substitute a linear expression into quadratic, cubic and product terms.

Name: _____ Suggested time: 10 minutes

Show your working. Substitute carefully and simplify.

01 Given y=3x-2

Write in terms of x and simplify

Simplify: $2x^3 - 2x^2 + 5y + 1$

02 Given y=5x-2

Write in terms of x and simplify

Simplify: $6x^2 + 4y + 6$

03 Given y=3x

Write in terms of x and simplify

Simplify: $3x^2 + 5xy + 5y - 4$

04 Given y=3x

Write in terms of x and simplify

Simplify: $4x^2 + 3xy + 3y + 2$

Answer sheet

Quadratic and cubic expressions | 4 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 $2x^3-2x^2+15x-9$

02 $6x^2+20x-2$

03 $18x^2+15x-4$

04 $13x^2+9x+2$

Worked solutions

Quadratic and cubic expressions

01 Given y=3x-2

$2x^3 -2x^2 +5y +1$

Substitute y=3x-2.

$2x^3 -2x^2 +5(3x-2) +1$

$2x^3 -2x^2 +15x -10 +1$

$2x^3-2x^2+15x-9$

02 Given y=5x-2

$6x^2 +4y +6$

Substitute y=5x-2.

$6x^2 +4(5x-2) +6$

$6x^2 +20x -8 +6$

$6x^2+20x-2$

Worked solutions

Quadratic and cubic expressions

03 Given y=3x

$3x^2 + 5xy + 5y - 4$

Substitute y=3x.

$3x^2 + 5x(3x) + 5(3x) - 4$

$3x^2 + 15x^2 + 15x - 4$

$(3+15)x^2 + 15x - 4$

$18x^2+15x-4$

04 Given y=3x

$4x^2 + 3xy + 3y + 2$

Substitute y=3x.

$4x^2 + 3x(3x) + 3(3x) + 2$

$4x^2 + 9x^2 + 9x + 2$

$(4+9)x^2 + 9x + 2$

$13x^2+9x+2$